

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_random_replay.
- Best TSPLIB holdout gap: tsplib_random_replay.
- Best synthetic holdout gap: tsplib_random_replay.

Run Metadata

- run_name: run_20260513_142231_k
- started_at_local: 2026-05-13 14:22:31
- finished_at_local: 2026-05-13 14:32:09
- duration_hhmm: 00:10
- duration_seconds: 578.242
- seed_offset: 10000
- replicate_label: k
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.767125	0.56	-0.032465
tsplib_random_replay	random	score_only	False	0.182623	0.0	0.116215	0.781271	0.76	0.010072
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.861375	0.56	-0.028913
tsplib_failure_replay_compression	failure	novelty_gate	True	0.213387	0.064916	0.159398	0.0	0.56	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.767125, and final complexity 0.56.
- Adaptation efficiency -0.032465 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.182623, synthetic 0.0, combined 0.116215.
- Accepted-epoch count 2, mean accepted code novelty 0.781271, and final complexity 0.76.
- Adaptation efficiency 0.010072 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.182623 across 7 instances; family means: ch=0.130649, kroD=0.25162, pcb=0.229765, pr=0.223199, rd=0.205815, st=0.106667.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 10118, "family": "berlin", "name": "berlin52", "optimality_gap": 0.341554}, {"best_known_cost": 2579, "cost": 3293, "family": "a", "name": "a280", "optimality_gap": 0.276851}, {"best_known_cost": 629, "cost": 773, "family": "eil", "name": "eil101", "optimality_gap": 0.228935}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.861375, and final complexity 0.56.
- Adaptation efficiency -0.028913 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.

- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
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- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Suite Results

Condition	Replay Mode	Final T SPLIB Gap	Final Sy nthetic Gap	Final Tr ansfer Gap	Code Novelty (mean)	Co mp lexi ty	Notes on Transfer vs Novelty
tsplib_random_replay (best)	Random Replay	**0.1826**	**0.0**	**0.1162**	0.7813	0.76	Best transfer performance despite lower code novelty than failure replay conditions. Synthetic gap zero indicates strong generalization on synthetic holdout.
tsplib_no_replay	None	0.2134	0.0649	0.1594	0.7671	0.56	Higher transfer gap than random replay, with slightly lower novelty but better than failure replay compression on transfer.
tsplib_failure_replay	Failure Replay	0.2134	0.0649	0.1594	0.8614	0.56	High code novelty but transfer and TSPLIB gaps equal to no replay; no transfer gain despite novelty increase.
tsplib_failure_replay_compression	Failure Replay + Compression	0.2134	0.0649	0.1594	0.0	0.56	No code novelty with compression pressure; transfer performance matches no replay and failure replay, indicating no benefit from compression or novelty gating in transfer.

Conservative Interpretation

- **Transfer Evidence:**
- The **tsplib_random_replay** condition yields the best **held-out TSPLIB gap (0.1826)** and **synthetic holdout gap (0.0)***, outperforming no replay and failure replay modes.
- The **mean transfer gap (0.1162)** is the lowest under random replay, indicating improved generalization to new problem instances.
- No replay and failure replay variants show identical TSPLIB and transfer gaps (~0.2134 and ~0.1594 respectively), suggesting no transfer improvement in these conditions.
- **Code Novelty vs Transfer:**

- Failure replay conditions increase code novelty (~0.86 for failure replay, 0.0 with compression), but **do not improve transfer metrics** relative to no replay.
- Random replay shows **lower code novelty (0.78)** compared to failure replay but achieves **better transfer performance**.
- Compression-aware failure replay results in **zero** code novelty and no transfer improvement, indicating compression or novelty gating suppresses novelty without transfer gain.
- **Replay Modes:**
- **Random Replay** positively affects transfer despite somewhat reduced novelty compared to failure replay.
- **Failure Replay**, with or without compression, fails to yield transfer improvement, though failure replay shows the highest code novelty.
- **No Replay** shows intermediate novelty and transfer performance.
- **Complexity:**
- Random replay condition uses more complex solutions (0.76) than others (0.56), possibly contributing to transfer gains.

Conclusion

- **Random replay of TSPLIB instances is the best strategy for improving transfer performance** on both held-out real and synthetic benchmarks.
- This improvement occurs **despite reduced code novelty** compared to failure replay; hence, **lexical novelty does not equate to improved algorithmic transfer** here.
- Failure replay modes increase novelty significantly but do **not translate to better transfer gaps**, indicating novelty increase alone is insufficient for transfer.
- Compression-aware replay suppresses novelty without effect on transfer or held-out gaps, suggesting no advantage here.
- Therefore, prioritizing **optimality gap and transfer measures** supports random replay as the optimal replay mode in this suite.